PYASC V2 · CANNBENCH OFFICIAL-TASKS V1.1.1 · 950PR

Target provenance, diagnosis и corrective reruns

Финальный результат

40/40 в обоих corrective runs

Ноль anti-cheat failures; manual score 135.0845, skill-driven score 128.7255

Что именно было взято из pyasc targets

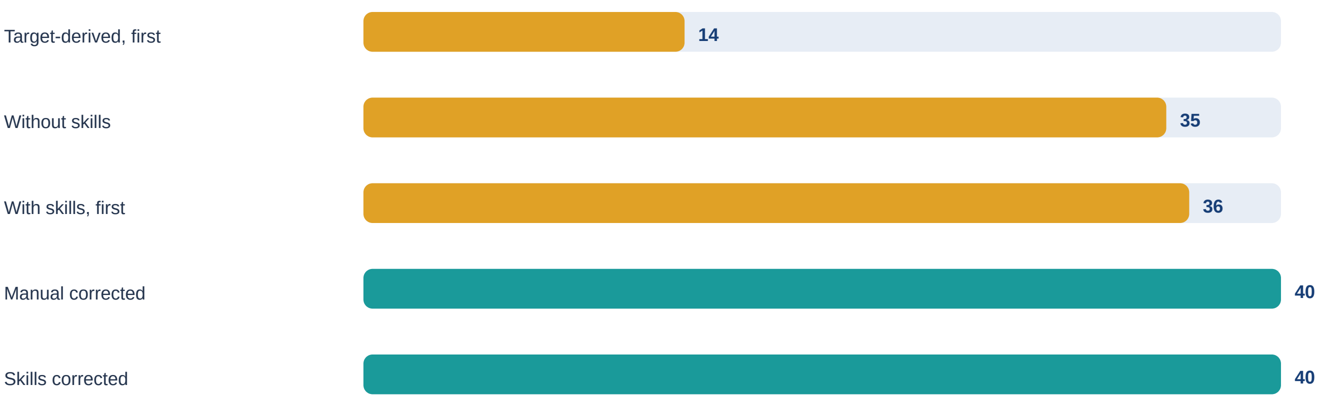
Первый «handwritten» arm не являлся буквальной копией текущего каталога `/home/aloschilov/workspace/pyasc-fork/python/test/asc2/target`. Исходники были извлечены из Git objects того же репозитория на commit `4d1db41d…` и адаптированы к CANNBench. Текущий `origin/v2` — `030e9b2c…` — уже использует `python/test/asctile/target`.

> **Корректная классификация:** target-derived adapter. Успешный ручной rerun — target-derived contract completion, а не verbatim target test.

Remote result matrix

Вариант	Job	Cases	Score	GMean	AC
Target-derived, first	job_79c05b96ed9d	14/40	69.794947	1.023126×	0
Without skills	job_75a7fee4ae6f	35/40	125.452428	0.616455×	0
With skills, first	job_ae3bfdefd087	36/40	124.607094	0.595558×	0
Manual corrected	job_a3e9153a2930	**40/40**	**135.084504**	0.525847×	0
Skills corrected	job_028f05fd8ec8	**40/40**	**128.725482**	0.457136×	0

Correctness: passed cases из 40



Per-operator result

Вариант	Оператор	Cases	Avg speedup	Score
Target first	Addcdiv	13/20	1.189408×	53.191518
Target first	GELU	1/20	0.880090×	16.603429
Without skills	Addcdiv	20/20	1.082186×	76.274837
Without skills	GELU	15/20	0.351157×	49.177591
Skills first	Addcdiv	20/20	0.920270×	72.679093
Skills first	GELU	16/20	0.385419×	51.928002
Manual corrected	Addcdiv	**20/20**	1.147209×	78.626238
Manual corrected	GELU	**20/20**	0.241033×	56.458266
Skills corrected	Addcdiv	**20/20**	0.917337×	72.599692
Skills corrected	GELU	**20/20**	0.227804×	56.125790

Partial-run GMean is not comparable as a complete-solution metric. Correctness is a hard gate.

Root cause and corrections

Target coverage gap

GELU target covered one FP32 approximation, not exact/BF16/wide ranges. Addcdiv fixed scalar=0.5 and bounded divisors. These tests were narrower than the CANNBench contract.

GELU

Stable erfc/Horner exact path and bounded-exp sigmoid tanh path remove negative-tail cancellation. Internal compute is FP32.

Addcdiv

FP32 internal division fixes unsupported BF16 division and scalar-zero FP16 NaN-position mismatch; one cast occurs at output.

Skill / UB

Exact tile 2048 used 795–820 KB versus a 253952-byte limit. The measured-safe tile 512 uses 199–205 KB. The skill now records this and requires adversarial camodel evidence.

Validation flow

Target references **source audit** → Manual / OpenCode **provenance** → Pinned-v2 QEMU **40/40 each** → 950PR camodel **passed** → CANNBench silicon **40/40 each**

Camodel adversarial suites passed dtype-specific tolerances and exact NaN/Inf positions. Cumulative suite ticks: manual GELU 40386 / Addcdiv 16201; skill GELU 38344 / Addcdiv 15277. Simulator ticks are not silicon speedups.

Worker provenance

- Accepted repair: Qwen session `ses_f9c952882ffeAnIEP58bPaIy7l`.
- Accepted review: Qwen session `ses_f9c8e48c5ffeFcOrr6Qvqf72Tj`.
- Both loaded repository-local `pyasc-cannbench-kernel`.
- GLM review `ses_f9c93e480ffeq5Hu4yhp750SLO` loaded the skill but timed out and was rejected.
- Accepted GELU SHA-256: `794e4ad018ba82487b36247eff5c3af5e1ee4925f384a09bb8d4a194a77fc1de`.

What CANNBench evaluates

CANNBench installs the self-contained wheel and runs exported operator functions on 950PR. It evaluates compile/runtime behavior, numerical correctness, anti-cheat, and case-level latency/speedup. It does not discover the skill-stack or generation history. “Without skills” versus “with skills” is established by local prompts, event logs, session IDs, native skill calls, and immutable hashes. See the leaderboard.

Next performance iteration

- GELU exact: compare tile 512/unroll2 with 1024/unroll1 while preserving 20/20.
- Addcdiv: add size-dispatched tile 2048; manual 1.147209× versus skill 0.917337× is the clearest tuning gap.
- Compare only complete, correctness-passing runs using case-level silicon latency.
- Keep source and runtime commits explicit during the upstream asc2→asctile transition.

Artifacts: initial diagnosis, corrective manifest, QEMU reports, camodel JSON, remote payloads/logs, immutable ZIPs, and worker traces are stored under `integrations/cannbench/comparisons/target-intersection-20260902/`.